

مکمل الفیرست

biology 102

chapter 20

DNA Tools

and

Biotechnology



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Concept 20.1

- 1) Which of the following processes helps bacterial cells protect their own DNA from restriction enzymes (endonucleases)?
- A) adding methyl groups to adenines and cytosines
 - B) using DNA ligase to seal the bacterial DNA into a closed circle
 - C) adding histones to protect the double-stranded DNA
 - D) forming "sticky ends" of bacterial DNA to prevent the enzyme (endonuclease) from attaching

Answer: A

2) What is the most logical sequence of steps for splicing foreign DNA into a plasmid and inserting the plasmid into a bacterium?

- I. Transform bacteria with a recombinant DNA molecule.
- II. Cut the plasmid DNA using restriction enzymes (endonucleases).
- III. Extract plasmid DNA from bacterial cells.
- IV. Hydrogen-bond the plasmid DNA to non-plasmid DNA fragments.
- V. Use ligase to seal plasmid DNA to non-plasmid DNA.

- A) II, III, V, IV, I
- B) III, II, IV, V, I
- C) III, IV, V, I, II
- D) IV, V, I, II, III

Answer: B

3) A principal problem with inserting an unmodified mammalian gene into a plasmid and then getting that gene expressed in bacteria is that _____.

- A) prokaryotes use a different genetic code from that of eukaryotes
- B) bacteria translate only mRNAs that have multiple messages
- C) bacteria cannot remove eukaryotic introns
- D) bacterial RNA polymerase cannot make RNA complementary to mammalian DNA

Answer: C

4) Why are yeast cells frequently used as hosts for cloning?

- A) They easily form colonies.
- B) They can remove exons from mRNA.
- C) They do not have plasmids.
- D) They are eukaryotic cells.

Answer: D

5) Why is sequencing an entire genome, such as that of *C. elegans*, a nematode, important for genetic research?

A) It allows researchers to use the sequence to build a "better" nematode, which is resistant to disease.

B) It allows research on a group of organisms we do not usually care much about.

C) A sequence that is found to have a particular function in the nematode is likely to have a closely related function in vertebrates.

D) A sequence that is found to have no introns in the nematode genome is likely to have acquired the introns from higher organisms.

Answer: C

6) Which of the following methods would be most successful in attempting to introduce a particular piece of DNA into an animal cell?

A) electroporation followed by recombination

B) introducing a plasmid into the cell

C) infecting the mouse cell with a Ti plasmid

D) transcription and translation

Answer: A

7) Why is it so important to be able to amplify DNA fragments when studying genes?

A) Before amplification, DNA fragments are likely to bind to RNA and would no longer be able to be analyzed.

B) A gene may represent only a millionth of the cell's DNA.

C) Restriction enzymes (endonucleases) cut DNA into fragments that are too small.

D) A clone requires multiple copies of each gene per clone.

Answer: B

8) *Pax-6* is a gene that is involved in eye formation in many invertebrates, such as *Drosophila*.

Pax-6 is also found in vertebrates. A *Pax-6* gene from a mouse can be expressed in a fly and the protein (PAX-6) leads to a compound fly eye. This information suggests which of the following characteristics of this gene?

- A) *Pax-6* genes are identical in nucleotide sequence.
- B) PAX-6 proteins have identical amino acid sequences.
- C) *Pax-6* is highly conserved and shows shared evolutionary ancestry.
- D) PAX-6 proteins are different for formation of different kinds of eyes.

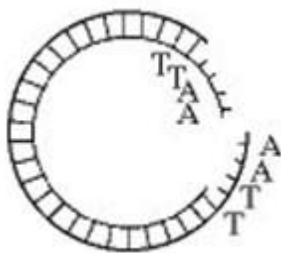
Answer: C

9) Which of the following characteristics of *Taq* polymerase make it useful in the PCR process?

- A) It is heat stable and can withstand the heating step of PCR.
- B) Only minute amounts are needed for each cycle of PCR.
- C) It binds more readily than other polymerases to the primers.
- D) It has regions that are complementary to the primers.

Answer: A

10) Use the figure to answer the following question



Which of the following enzymes was used to produce the molecule of DNA in the figure?

- A) ligase
- B) a restriction enzyme (endonuclease)
- C) RNA polymerase
- D) DNA polymerase

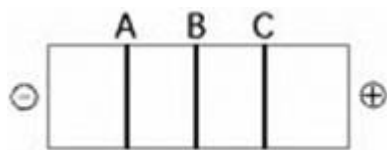
Answer: B

11) Use the figure to answer the following question.

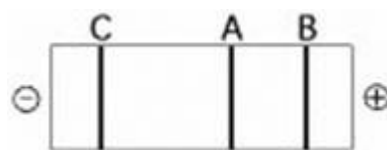


The segment of DNA shown in the figure has restriction sites I and II, which create restriction fragments A, B, and C. Which of the gels produced by electrophoresis best represents the separation and identity of these fragments?

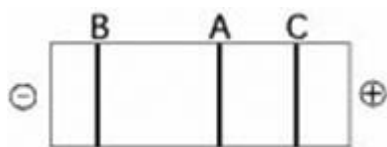
A)



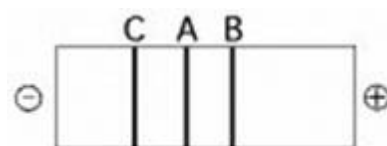
B)



C)



D)



Answer: B

12) For which of the following processes can dideoxynucleotides be used?

- A) to separate different sized DNA fragments
- B) to produce cDNA from mRNA
- C) to sequence a DNA fragment
- D) to visualize DNA expression

Answer: C

13) Many identical copies of genes cloned in bacteria are produced as a result of which of the following processes?

- A) plasmid replication
- B) bacterial cell reproduction
- C) transformation
- D) plasmid and bacterial cell reproduction

Answer: D

14) Which of the following sequences is most likely to be cut by a restriction enzyme?

- A) 5'-AATTCT 3'
3'-TTAAGA-5'
- B) 5'-AATATT-3'
3'-TTATAA-5'
- C) 5'-AAAATT-3'
3'-TTTAA-5'
- D) 5'-ACTACT-3'
3'-TGATGA-5'

Answer: B

15) What information is critical to the success of polymerase chain reaction (PCR) itself?

- A) The DNA sequence of the ends of the DNA to be amplified must be known.

B) The complete DNA sequence of the DNA to be amplified must be known.

C) The sequence of restriction-enzyme recognition sites in the DNA to be amplified must be known.

D) The sequence of restriction-enzyme recognition sites in the DNA to be amplified and in the plasmid where the amplified DNA fragment will be cloned must be known.

Answer: A

16) Which of the following correctly lists the processes in order for one cycle of polymerase chain reaction (PCR)?

A) denature DNA; add fresh enzyme; anneal primers; add dNTPs; extend primers

B) anneal primers; denature DNA; extend primers

C) extend primers; anneal primers; denature DNA

D) denature DNA; anneal primers; extend primers

Answer: D

17) The final step in a Sanger DNA sequencing reaction is to run the DNA fragments on a gel.

What purpose does this serve?

A) It adds ddNTP to the end of each DNA fragment.

B) It changes the length of the DNA fragments.

C) It separates DNA fragments based on their charge.

D) It separates DNA fragments generated during the sequencing reaction based on onenucleotide differences in their size.

Answer: D

Concept 20.2

18) How can a gene that contains introns be made shorter (but remain functional) for genetic engineering purposes?

A) using a restriction enzyme (endonuclease) to cut the gene into shorter pieces

B) using reverse transcriptase to reconstruct the gene from its mRNA

C) using DNA polymerase to reconstruct the gene from its polypeptide product

D) using DNA ligase to put together fragments of the DNA that code for a particular polypeptide

Answer: B

19) Which of the following enzymes is required to make complementary DNA (cDNA) from RNA?

A) restriction enzymes (endonucleases)

B) helicase

C) DNA ligase

D) reverse transcriptase

Answer: D

20) How have DNA microarrays made a huge impact on genomic studies?

A) They can be used to eliminate the function of any gene in the genome.

B) They can be used to introduce entire genomes into bacterial cells.

C) They allow the expression of many or even all of the genes in the genome to be compared at once.

D) They allow physical maps of the genome to be assembled in a very short time.

Answer: C

21) RNAi methodology uses double-stranded pieces of RNA to trigger breakdown of a specific mRNA or inhibit its translation. For which of the following processes might this technique be useful?

- A) to decrease the production from a harmful mutated gene
- B) to destroy an unwanted allele in a homozygous individual
- C) to form a knockout organism that will not pass the deleted sequence to its progeny
- D) to raise the concentration of a desired protein

Answer: A

22) Which of the following statements correctly describes how RNA interference (RNAi) silences selected genes?

- A) Synthetic double-stranded RNA molecules stop transcription from occurring.
- B) Synthetic double-stranded RNA molecules stop DNA replication from occurring.
- C) Synthetic double-stranded RNA molecules trigger the breakdown of a gene's messenger RNA.
- D) Synthetic double-stranded RNA molecules stop mitosis from occurring.

Answer: C

23) In large scale, genome-wide association studies in humans, what types of genetic markers do researchers look for?

- A) lengthy sequences that might be shared by most members of a population
- B) SNPs where one allele is found more often in persons with a particular disorder than in healthy controls

- C) SNPs where one allele is found in families with a particular introns sequence
- D) SNPs where one allele is found in two or more adjacent genes

Answer: B

24) For a particular microarray assay (DNA chip), cDNA has been made from the mRNAs of a dozen patients' breast tumor biopsies. Which of the following types of evidence will researchers be looking for in order to determine if the cells are cancerous?

- A) a particular gene that is amplified in all or most of the patient samples
- B) a pattern of fluorescence that indicates which cells are over proliferating
- C) a pattern shared among some or all of the samples that indicates gene expression differing from control samples
- D) a group of cDNAs that match those in non-breast cancer control samples from the same population

Answer: C

25) In a comparison of two DNA sequences found in the same location on homologous chromosomes, one of the homologs carries the sequence 5'-AACTACGA-3', and the other homolog carries the sequence 5'-AACTTCGA-3'. Within a population, you discover that each of these sequences is common.

Which of the following statements correctly describes these sequences?

- A) They contain a SNP that may be useful for genetic mapping.

- B) They can identify a protein-coding region of a gene.
- C) They may cause disease.
- D) They may carry out RNA interference.

Answer: A

26) Which of the following processes uses labeled probes to visualize the expression of genes in whole tissues and organisms?

- A) RT-PCR
- B) *in situ* hybridization
- C) DNA microarrays
- D) RNA interference

Answer: B

Concept 20.3

27) Which of the following processes is most like the formation of identical twins?

- A) cell cloning
- B) therapeutic cloning
- C) use of adult stem cells
- D) organismal cloning

Answer: D

28) In 1997, Dolly the sheep was cloned. Which of the following processes was used?

- A) replication and dedifferentiation of adult stem cells from sheep bone marrow
- B) separation of an early stage sheep blastula into separate cells, one of which was incubated in a surrogate ewe
- C) transfer of an adult cell's nucleus into an enucleated sheep egg, followed by incubation in a surrogate

D) isolation of stem cells from a lamb embryo and production of a zygote equivalent

Answer: C

29) Which of the following problems with animal cloning might result in premature death of the clones?

A) use of pluripotent instead of totipotent stem cells

B) use of nuclear DNA as well as mtDNA

C) abnormal gene regulation due to variant methylation

D) the indefinite replication of totipotent stem cells

Answer: C

30) Reproductive cloning of human embryos is generally considered unethical. However, on the subject of therapeutic cloning, there is a wider divergence of opinion. Which of the following statements provides a likely explanation?

A) The use of adult stem cells is likely to produce more cell types than the use of embryonic stem cells.

B) Cloning to produce embryonic stem cells may lead to great medical benefits for many.

C) Cloning to produce stem cells relies on a different initial procedure than reproductive cloning.

D) A clone that lives until the blastocyst stage does not yet have human DNA.

Answer: B

31) Which of the following statements correctly describes one of the main differences between embryonic stem cells and adult stem cells?

A) Embryonic stem cells only differentiate into only eggs and sperm, and adult stem cells differentiate into any type of cell.

B) Embryonic stem cells can give rise to all cell types in the organism, and adult stem cells cannot.

C) Embryonic stem cells can continue to reproduce for an indefinite period, and adult stem cells cannot.

D) One aim of using embryonic stem cells is to provide cells for repair of diseased tissue.

Answer: B

32) A researcher is using adult stem cells and comparing them to other adult cells from the same tissue. Which of the following findings is likely from this comparison?

A) The cells from the two sources exhibit different patterns of DNA methylation.

B) Adult stem cells have more DNA nucleotides than their counterparts.

C) The two kinds of cells have virtually identical gene expression patterns in microarrays.

D) The non-stem cells have fewer repressed genes.

Answer: A

33) In animals, what is the difference between reproductive cloning and therapeutic cloning?

A) Reproductive cloning uses totipotent cells, whereas therapeutic cloning does not.

B) Reproductive cloning uses embryonic stem cells, whereas therapeutic cloning does not.

C) Therapeutic cloning uses nuclei of adult cells transplanted into enucleated unfertilized eggs.

D) Therapeutic cloning supplies cells for repair of diseased or injured organs.

Answer: D

34) The first cloned cat, called Carbon Copy, was a calico, but she looked significantly different from her female parent. Why was this the case with this clone?

A) The cloning was done poorly and it was likely that some contaminating cat DNA became part of Carbon Copy's genome.

B) Fur color genes in cats is determined by differential acetylation patterns.

C) Cloned animals have been found to have a higher frequency of transposon activation.

D) X chromosome inactivation in the embryo is random and produces different patterns.

Answer: D

35) In recent times, it has been shown that adult cells can be induced to become pluripotent stem cells (iPS). To make this conversion, what has been done to the adult cells?

A) A retrovirus is used to introduce four specific master regulatory genes.

B) The adult stem cells must be fused with embryonic cells.

C) Cytoplasm from embryonic cells is injected into the adult cells.

D) The nucleus of an embryonic cell is used to replace the nucleus of an adult cell.

Answer: A

36) Suppose that a researcher is successful at producing induced pluripotent stem cells (iPS) for replacement of pancreatic insulin-producing cells for people with type 1 diabetes. Which of the following could still pose problems for this individual?

I. the possibility that, once introduced into the patient, the iPS cells produce non-pancreatic cells

II. the failure of the iPS cells to take up residence in the pancreas

III. the inability of the iPS cells to respond to appropriate regulatory signals

A) I only

B) II only

C) III only

D) I, II, and III

Answer: D

Concept 20.4

37) Which of the following statements correctly describes one characteristic of genetically engineered plants?

A) They are more difficult to develop than genetically engineered animals.

B) They include transgenic rice plants that can grow in water more salty than seawater.

C) They are used in research but not yet in commercial agricultural production.

D) They are banned throughout the world.

Answer: B

38) Scientists developed a set of guidelines to address the safety of DNA technology. Which of the following safety measures is one that has been adopted by researchers?

A) Microorganisms used in recombinant DNA experiments must be genetically crippled to

ensure that they cannot survive outside of the laboratory.

B) Genetically modified organisms are not allowed to be part of our food supply.

C) Transgenic plants are engineered so that the plant genes cannot hybridize.

D) Experiments involving HIV or other potentially dangerous viruses have been banned.

Answer: A

39) Which of the following statements describes one of the technical reasons why gene therapy is problematic in humans?

A) Most cells with an engineered gene do not produce gene product.

B) Cells with transferred genes are unlikely to replicate.

C) Transferred genes may not have appropriately controlled activity.

D) mRNA from transferred genes cannot be translated.

Answer: C

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Answer: C

40) One possible use of transgenic plants is in the production of human proteins, such as

vaccines. Which of the following issues is a possible hindrance that must be overcome in order for this process to work properly?

- A) prevention of transmission of plant allergens to the vaccine recipients
- B) prevention of vaccine-containing plants being consumed by insects
- C) use of plant cells to translate non-plant-derived mRNA
- D) inability of the human digestive system to accept plant-derived protein

Answer: A

41) Which of the following characteristics of short tandem repeats (STRs) DNA makes it useful for DNA fingerprinting?

- A) The number of repeats is highly variable from person to person or animal to animal.
- B) The sequence of DNA that is repeated varies significantly from individual to individual.
- C) The sequence variation is acted upon differently by natural selection in different environments.
- D) Every racial and ethnic group has inherited different short tandem repeats.

Answer: A

42) Transgenic mice are useful to human researchers because they _____.

- A) can be valuable animal models of human disease
- B) are essential for mapping human genes
- C) are now used in place of bacteria for cloning human genes
- D) were instrumental in pinpointing the location of the *huntingtin* gene

Answer: A

43) Which of the following information or processes does gene therapy require?

A) the knowledge and availability of the defective gene but not the normal allele

B) the ability to introduce the normal allele into the patient

C) the ability to express the introduced gene at any time in the tissue site within the patient

D) the ability to introduce a defective allele into a patient

Answer: B

44) What is the most favorable characteristic of retroviruses that makes them useful in gene therapy applications?

A) Retroviruses have an RNA genome.

B) Retroviruses possess reverse transcriptase.

C) DNA copies of retroviral genomes become integrated into the genome of the infected cell.

D) Retroviruses mutate often.

Answer: C

45) Why might using retroviral vectors for gene therapy increase the patient's risk of developing cancer?

A) Retroviral vectors may introduce proteins from the virus.

B) Retroviral vectors do not express the genes that were introduced into a patient's cells.

C) Retroviral vectors do not integrate their recombinant DNA into the patient's genome.

D) Retroviral vectors integrate recombinant DNA into the genome in ways that may misregulate the expression of genes at or near the site of integration.

Answer: D

46) In the form of gene therapy used successfully for severe combined immunodeficiency syndrome, SCID-X1, how is the genetic engineering of human cells achieved?

A) by injecting engineered viruses into the patient's bloodstream

B) by injecting engineered viruses into the patient's bone marrow

C) by treating a relative's cultured bone marrow cells with genetically engineered viruses and then injecting these cells into the patient's bone marrow

D) by isolating the patient's bone marrow cells, infecting them with genetically engineered viruses, and injecting them back into the patient's bone marrow

Answer: D

47) One predicted aspect of climate change is that climates, including precipitation and temperature, over most of Earth will become more variable. Which of the following is a good

strategy for genetically engineering crops if this is true?

A) Use only plant crops that are genetically engineered.

B) Genetically engineer most crops to withstand very long droughts.

C) Genetically engineer several genotypes within single crop types.

D) Genetically engineer the fastest growing crops possible.

Answer: C

